A Deep Learning Framework for Functional Eutrophication Assessment: Mapping the Trophic Transfer Index

Authors: Adolf Stips*, Ove Pärn, Luca Polimene, Diego Macias-Moy*, Svetla Miladinova, Elisa Garcia-Gorriz, Olaf Duteil

Journal: *Environmental Modelling & Software* (2026)

Contact: `adolf.stips@ext.ec.europa.eu` | `adolf.stips@gmail.com`

Overview

This repository contains the complete data-processing, neural network training, Explainable AI (SHAP), and visualization pipeline for the manuscript **"A Deep Learning Framework for Functional Eutrophication Assessment: Mapping the Trophic Transfer Index"**.

As a synthetic proof-of-concept, this framework applies Multi-Layer Perceptron (`MLP`) and Long Short-Term Memory (`LSTM`) neural networks to emulate the continuous **Trophic Transfer Index (TTI)** and classify **Good Environmental Status (GES; TTI > 0.71)** across the Baltic Sea (29,502 marine grid cells) using 70 years (1953–2022; 14 pentads) of simulated physical-biogeochemical data from the coupled **GETM-ERGOM** ecosystem model.

The repository evaluates four core experimental configurations across three environmental variable suites (`Comprehensive`, `MSFD_Baseline`, and `Remote_Sensing`):

1. **`MLP_SPAT`**: Feedforward spatial gap-filling via non-overlapping 10-Fold Spatial Block Cross-Validation (10×10 pixel blocks; 80% Train / 10% Validation / 10% Test rotation).

2. **`LSTM_SPAT`**: Recurrent spatial gap-filling ingesting 35-year (7-pentad) chronological sequences (`[batch, sequence_length = 7, features]`) via 10-Fold Spatial Block Cross-Validation.

3. **`MLP_TIME`**: Feedforward rolling-origin temporal hindcast across 6 unseen target pentads (1995–2020) partitioned across non-overlapping spatial blocks (60% Train / 20% Val / 20% Test).

4. **`LSTM_TIME`**: Recurrent rolling-origin temporal hindcast ingesting 35-year lookback sequences across 6 unseen target pentads (1995–2020).

Repository Structure

```text
TTI_ANN_FIG/
├── README.md                    # Workflow documentation and instructions
├── LICENSE                      # GNU General Public License v3.0 (GPL-3.0)
├── GES_Baltic_5year_ALLE_Combined_4D.nc.zip   # Zipped NetCDF input dataset (14 pentads, 1955-2020)
│
├── 00_Run_All_Workflow.R              # Master script to execute the full pipeline
├── 01_Kfold_MLP_SPAT.R                # 10-Fold Spatial Block CV (MLP_SPAT)
├── 02_Kfold_LSTM_SPAT.R               # 10-Fold Spatial Block CV (LSTM_SPAT)
├── 03_Rolling_MLP_TIME.R              # Rolling-Origin Temporal Forecast (MLP_TIME)
├── 04_Rolling_LSTM_TIME.R             # Rolling-Origin Temporal Forecast (LSTM_TIME)
├── 05_Spatial_Ensemble_MLP_SPAT.R     # 10-Fold Spatial Bootstrap Ensemble Reconstruction
├── 06_Generate_Manuscript_Tables.R    # Generates Manuscript Tables 2, 3, and 4
├── 07_Plot_Fig1_Spatial_Block_Map.R   # Generates Figure 1
├── 08_Plot_Figs2_3_4_SHAP_Analysis.R  # Generates Figures 2, 3, and 4
├── 09_Plot_Figs5_6_S3_Spatial_Ensemble.R  # Generates Figures 5, 6, and Supplementary Fig S3
├── 10_Plot_FigsS1_S2_Confusion_Matrices.R  # Generates Supplementary Figures S1 and S2
│
├── Output/                      # Pre-computed & generated .rds model outputs
│   ├── Summary_MLP_SPAT.rds
│   ├── Summary_LSTM_SPAT.rds
│   ├── Summary_MLP_TIME.rds
```

```
│   ├── Summary_LSTM_TIME.rds
│   ├── SHAP_RAW_MLP_SPAT.rds
│   ├── SHAP_RAW_LSTM_SPAT.rds
│   ├── SHAP_RAW_MLP_TIME.rds
│   ├── SHAP_RAW_LSTM_TIME.rds
│   ├── CF_Matrix_MLP_SPAT.rds
│   ├── CF_Matrix_LSTM_SPAT.rds
│   ├── CF_Matrix_MLP_TIME.rds
│   ├── CF_Matrix_LSTM_TIME.rds
│   ├── Ensemble_Map_Data_MLP_SPAT.rds
│   └── Ensemble_TimeSeries_MLP_SPAT.rds
│
├── Tables/                    # Generated Manuscript Tables (.docx and .csv)
└── Figures/                   # Generated Publication Figures (.png, 500 DPI)
```

System Requirements & Installation

Hardware & Software Environment

- **R Version:** 4.4.2 (or newer)
- **LibTorch / GPU Acceleration:** torch (v0.16.3), tested with NVIDIA CUDA 12.8 and cuDNN 9.6.7 (scripts automatically fall back to "cpu" if a CUDA GPU is unavailable).

Required R Packages

Install all required CRAN dependencies prior to running the workflow:

```
install.packages(c(
  "torch", "terra", "dplyr", "abind", "fastshap", "shapviz",
  "Metrics", "cvms", "tibble", "gt", "ggplot2", "ggpubr",
  "patchwork", "scales", "sf", "rnaturalearth", "rnaturalearthdata"
))
```

Reproducing the Workflow

Option A: Fast Regeneration of Tables and Figures (No Retraining Required)

All pre-computed model summaries, SHAP distributions, ensemble reconstructions, and confusion matrices are provided in ./Output/. To regenerate all manuscript tables and figures in under 1 minute without retraining the neural networks:

1. Open 00_Run_All_Workflow.R and set RUN_TRAINING <- FALSE (Line 13).
2. Run:

source("00_Run_All_Workflow.R")

Option B: Complete End-to-End Reanalysis (Full Model Training + Post-Processing)

To unzip the raw NetCDF dataset, execute all 10-fold spatial cross-validations, rolling-origin temporal hindcasts, Monte Carlo SHAP approximations (nsim = 50), and regenerate all tables and figures from scratch:

1. Ensure RUN_TRAINING <- TRUE in 00_Run_All_Workflow.R.
2. Run:

source("00_Run_All_Workflow.R")

Script-to-Manuscript Mapping

Script	**Description**	**Manuscript Deliverable**
01_Kfold_MLP_SPAT.R	10-fold spatial block CV & SHAP (MLP_SPAT)	Table 2, Table 4, Figs 3, 4, S1
02_Kfold_LSTM_SPAT.R	10-fold spatial block CV & SHAP (LSTM_SPAT)	Table 2, Table 4, Figs 3, 4, S2
03_Rolling_MLP_TIME.R	30-year rolling-origin hindcast & SHAP (MLP_TIME)	Table 3, Table 4, Fig 2

Script	Description	Manuscript Deliverable
04_Rolling_LSTM_TIME.R	30-year rolling-origin hindcast & SHAP (LSTM_TIME)	Table 3, Table 4, Fig 2
05_Spatial_Ensemble_MLP_SPAT.R	10-fold full-basin Fisher-*z* ensemble reconstruction	Figs 5, 6, S3
06_Generate_Manuscript_Tables.R	Formats performance & GFI metrics into .docx & .csv	**Table 2, Table 3, Table 4**
07_Plot_Fig1_Spatial_Block_Map.R	Maps Fold 1 (10 × 10 blocks; 80/10/10 split)	**Figure 1**
08_Plot_Figs2_3_4_SHAP_Analysis.R	Plots temporal/spatial GFI stability & beeswarm plots	**Figure 2, Figure 3, Figure 4**
09_Plot_Figs5_6_S3_Spatial_Ensemble.R	Plots basin time series, spatial TTI maps & SD maps	**Figure 5, Figure 6, Figure S3**
10_Plot_FigsS1_S2_Confusion_Matrices.R	Plots aggregated global confusion matrices	**Figure S1, Figure S2**

Reproducibility & Zero-Leakage Guarantees

- **Deterministic Seeding:** All spatial block assignments, fold rotations, Torch weight initializations, and Monte Carlo SHAP sub-samplings are strictly controlled via fixed random seeds (set.seed(123) and torch_manual_seed(123)).
- **Zero-Leakage Standardisation:** Across every spatial fold and temporal rolling origin, predictor standardization parameters (μ, σ) are computed exclusively from the isolated training partition and subsequently applied to the validation and unseen test partitions.
- **Separation of Early Stopping and Evaluation:** All spatial and temporal experiments enforce a strict 3-way split (Train / Validation / Test), ensuring that dynamic early stopping (patience = 20) is triggered solely on the validation partition while all reported metrics and SHAP values are evaluated on the unseen test partition.

License

This project is licensed under the **GNU General Public License v3.0 (GPL-3.0)**.

[![License: GPL v3](https://img.shields.io/badge/License-GPLv3-blue.svg)](https://www.gnu.org/licenses/gpl-3.0)

[![R Version](https://img.shields.io/badge/R-4.4.2-276DC3.svg)](https://www.r-project.org/)

[![PyTorch](https://img.shields.io/badge/torch-0.16.3-EE4C2C.svg)](https://torch.mlverse.org/)